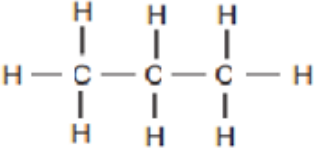
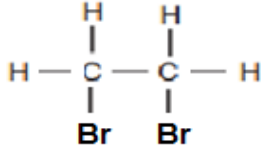
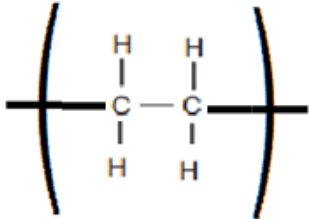
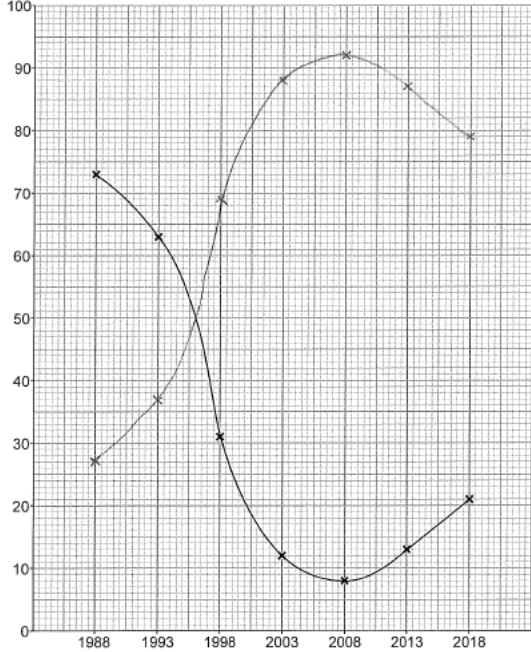


COMMON QUESTIONS

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7/1	(a)	(i)		D and E both needed, either order		1		1		
		(ii)		propene	1			1		
		(iii)			1			1		
	(b)	(i)		 (1)  (1)		2		2		
		(ii)		decolourises / goes colourless neutral answers – goes clear / changes colour	1			1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		(addition) polymerisation	1			1		
	(c)	(i)		0.15 (2) if answer incorrect award (1) for any of following $\frac{12}{100} \times 1.25$ 0.0125×12 $15 / 150000$		2		2	2	

Question				Marking details	Marks available																													
					AO1	AO2	AO3	Total	Maths	Prac																								
		(ii)		<table><caption>Data points from the graph</caption><tr><th>Year</th><th>Series 1 (Decreasing)</th><th>Series 2 (Increasing)</th></tr><tr><td>1988</td><td>73</td><td>27</td></tr><tr><td>1993</td><td>63</td><td>37</td></tr><tr><td>1998</td><td>31</td><td>69</td></tr><tr><td>2003</td><td>12</td><td>88</td></tr><tr><td>2008</td><td>8</td><td>92</td></tr><tr><td>2013</td><td>13</td><td>87</td></tr><tr><td>2018</td><td>21</td><td>79</td></tr></table> award (2) for all points plotted correctly – tolerance ± 1 square award (1) for any four correct points award (1) for curved line do not accept - point to point line	Year	Series 1 (Decreasing)	Series 2 (Increasing)	1988	73	27	1993	63	37	1998	31	69	2003	12	88	2008	8	92	2013	13	87	2018	21	79		3		3	3	
Year	Series 1 (Decreasing)	Series 2 (Increasing)																																
1988	73	27																																
1993	63	37																																
1998	31	69																																
2003	12	88																																
2008	8	92																																
2013	13	87																																
2018	21	79																																

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)	I	1996 accept 1995 / 1997 accept range 1995-1996 / 1996-1997		1		1		1
			II	2 : 8 1 : 4 4 : 1 8 : 2 20 : 80			1	1	1	
		(iv)		award (1) for any of following actions people use bag for life / alternative bags people reuse bags supermarkets charge for bags / stop giving free plastic bags government introduced a charge people have become more aware of issues to do with plastic waste people want to reduce amount of plastic going to landfill / incineration people want to reduce the amount of plastic pollution / microplastics			1	1		
				Question 7/1 total	4	9	2	15	6	2